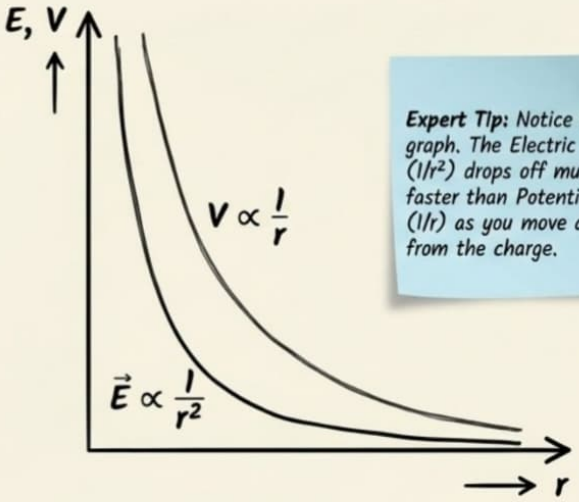
Electrostatic Potential & Capacitance

Potential Difference

Pot. diff. b/w 2 points in an $\vec{E}$ is defined as the amount of work done in moving a unit positive charge from one point to the other against the electrostatic forces.

Let us consider a point charge +q located at a point O. Let A & B be 2 points in its electric field. When a test charge q_0 is moved from A to B, a work W_{AB} has to be done in moving against the repulsive forces exerted by the charge +q. We then calculate the pot. diff. between 2 points A & B by the equation:

$$\boxed{V = V_A - V_B = \frac{W_{AB}}{q_0}}$$



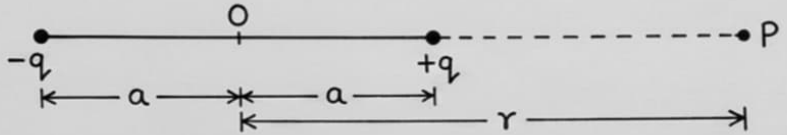
Expert Tip: Notice the graph. The Electric Field ($1/r^2$) drops off much faster than Potential ($1/r$) as you move away from the charge.

SI unit of pot. diff. is Volt named after Alessandro Volta.
$1V = 1JC^{-1}$

Therefore, Pot. diff. between 2 points in an $\vec{E}$ is said to be 1V if 1 J of work has to be done in moving a positive charge of 1 C from one point to other against the electrostatic forces.
It is a scalar quantity.

Potential Difference of a Dipole

* At axial point



Let us consider 2 point charges $+q$ & $-q$ separated by a distance $2a$, and let there be a point P at axis of dipole 'r' distance away from the center of dipole.

$\therefore$ Potential diff. due to $-q$:

$$V_{-q} = \frac{k(-q)}{r+a}$$

Similarly, pot. diff due to $+q$:

$$V_{+q} = \frac{kq}{r-a}$$

$$\therefore V_P = V_{-q} + V_{+q} = kq\left[\frac{-1}{(r+a)} + \frac{1}{(r-a)}\right]$$

$$= V_P = k\cdot q\left[\frac{-r+a+r+a}{(r+a)(r-a)}\right]$$

$$= V_P = k\cdot q\left[\frac{2a}{r^2-a^2}\right] = \frac{2aq\cdot k}{r^2-a^2}$$

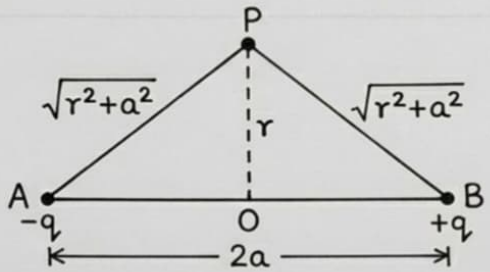
$$\boxed{\therefore V_P = V_{axial} = \frac{P\cdot k}{r^2-a^2}}$$

If $r \gg a$, $a^2 \approx 0$.

$$\boxed{\therefore V_{axial} = \frac{kP}{r^2}}$$

Potential Difference of a Dipole

* At equatorial point



Symmetry in action!
Any point on the equatorial plane is equidistant from both charges, making the net work to bring a charge from infinity exactly zero.

Let us consider 2 point charges $-q$ & $+q$ separated by a distance $2a$ & let there be a point $P \perp AB$ at pt. O which is at the mid of dipole. Let $OP = r$.
Let distance between P & $-q$ = dist. b/w P & $+q$ = $\sqrt{r^2+a^2}$

$\therefore$ Potential difference due to $-q$:

$$V_{-q} = \frac{k(-q)}{\sqrt{r^2+a^2}}$$

Similarly, potential due to $+q$:

$$V_{+q} = \frac{k(+q)}{\sqrt{r^2+a^2}}$$

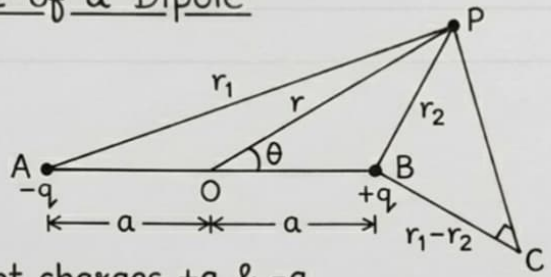
$$V_P = V_{-q} + V_{+q}$$

or $$V_P = \frac{-k \cdot q}{\sqrt{r^2+a^2}} + \frac{k \cdot q}{\sqrt{r^2+a^2}} = 0$$

$$\boxed{\therefore V_{equatorial} = 0}$$

Potential Difference of a Dipole

* General point



The Universal Equation: Plug in $\theta = 0°$ for axial, or $\theta = 90°$ for equatorial, and watch watch it match our previous slides perfectly!

Let us consider 2 point charges $+q$ & $-q$ at a dist. $2a$ & let P be a point outside the dipole r dist from O such that OP makes an angle θ with AB.

Let $AP = r_1$ & $BP = r_2$, where A & B are points at charges $-q$ & $+q$ resp. $r_1 > r_2$.

Let there be a line BC falling at AP such that $CP = BP = r_2$ $\therefore AC = r_1 - r_2$

In ΔACB,

$$\cos\theta = \frac{AC}{AB} = \frac{r_1 - r_2}{2a} \quad \therefore 2a \cdot \cos\theta = r_1 - r_2$$

Also, $r_1 \cdot r_2 \approx r^2$

$\therefore$ Pot diff due to charge $-q$ at P,

$$V_{-q} = \frac{k(-q)}{r_1}$$

Similarly, pot diff at P due to $+q$,

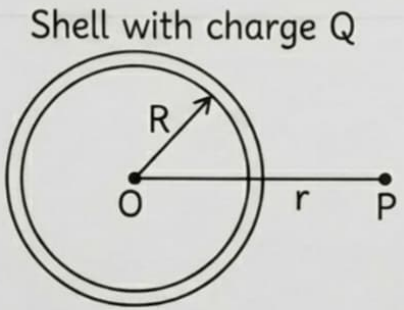
$$V_{+q} = \frac{kq}{r_2}$$

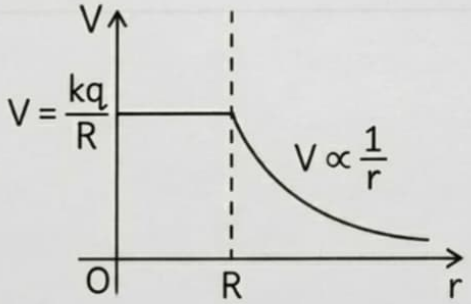
$$\therefore V_P = V_{-q} + V_{+q} = \frac{-kq}{r_1} + \frac{kq}{r_2} = kq\left[\frac{1}{r_2} - \frac{1}{r_1}\right]$$

$$V_P = kq\left[\frac{r_1 - r_2}{r_1 \cdot r_2}\right] = kq\left[\frac{2a\cos\theta}{r^2}\right] = \frac{2aq \cdot k \cdot \cos\theta}{r^2}$$

$$\boxed{\therefore V_{general} = \frac{kP \cdot \cos\theta}{r^2}}$$

Electric potential due to a uniformly charged thin spherical shell:





1. When pt. P lies outside the shell (r > R)

$$V = \frac{kq}{r}$$

2. When pt. P lies on the surface of shell (r = R)

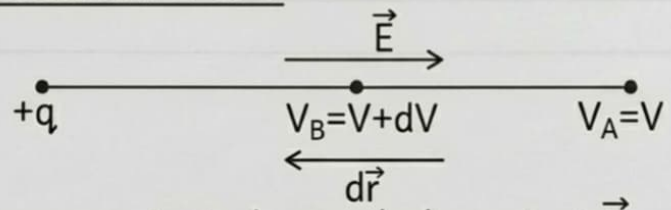
$$V = \frac{kq}{R}$$

3. When pt. P lies inside the shell (r < R)

$$V = \frac{kq}{R}$$

Notice how the potential inside the shell is constant and equals the surface potential. This is because the electric field inside a hollow conductor is zero, so no work is required to move a charge within it.

Relationship between $\vec{E}$ & V



The 'Potential Gradient'. The negative sign indicates that the electric field always points in the direction of decreasing potential.

Force reqd to move a test charge (q_0) against $\vec{E}$,

$\therefore \vec{F} = -q_0\vec{E}$

Now, work done to take test charge from A to B,

$W = F\cdot d\vec{r}$ or $W = -q_0\cdot E\cdot d\vec{r}$ --- (I)

Also, work done is given as;

$W = q_0\,[\Delta V]$

or $W = q_0\,[V_B - V_A]$

or $W = q_0\,[V + dV - V]$

or $W = q_0\cdot dV$ --- (II)

SI unit of EF = V/m or Vm^{-1}.

Equating (I) & (II), we get:

$-q_0\,E\,dr = q_0\,dV$

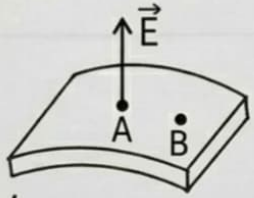
$$\boxed{\therefore E = \frac{-dV}{dr}}$$

Also, $V = -\int_{r_1}^{r_2}\vec{E}\cdot d\vec{r}$

If we take r_1 from ∞, $V = -\int_{\infty}^{r}\vec{E}\cdot d\vec{r}$

Equipotential surfaces & their properties

1. No work is done in moving a test charge over an equipotential surface.



$$W_{AB} = q_0\Delta V = q_0(V_B - V_A)$$

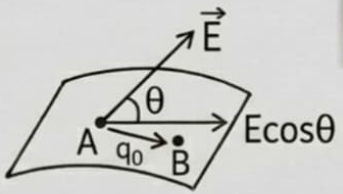
Since, on equipotential surface, $V_A = V_B = V$

$$\therefore W_{AB} = q_0(V-V) = q_0 \cdot 0 = 0$$

$$\boxed{\therefore W = 0}$$

Key Concept: Because no work is done moving the surface, the force (and thus the electric field) must be perpendicular to the direction of motion (the surface itself).

2. EFL are always normal to the equipotential surface at every point.



∵ If the field were not normal to the equipotential surface, it would have a non-zero component along the surface (E cosθ). So, to move a test to move test charge (q_0) against this component, work has to be be done. But there is no p.d. between 2 pts in an equipotential surface, & thus no work is required to be done to move a test charge.

∴ EFL ⊥ equipotential surface.

Properties of Equipotential Surfaces (cont.)

3. Equipotential surfaces are closer together in the regions of weak fields.

$$\vec{E} \propto \frac{1}{dr \text{ (separation)}}$$

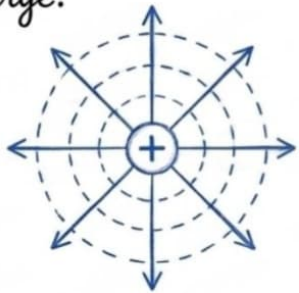
Expert Correction: Wait! The text says "weak fields" but the math $E \propto \frac{1}{dr}$ tells the true story. A smaller separation (dr) creates a LARGER E. So equipotential surfaces are actually closer together in regions of STRONG fields!

4. No two equipotential surfaces can intersect each other.

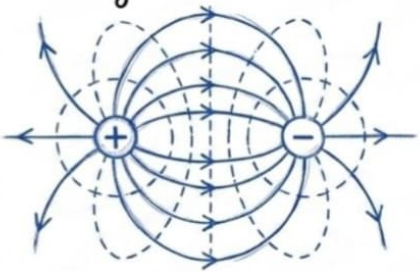
   ∴ because If they intersect, then there will be 2 values of potential at the point of intersection, which is impossible.

Equipotential surfaces of various charge systems

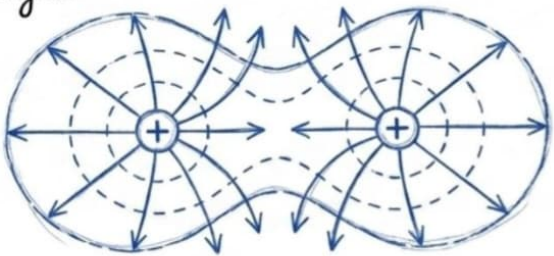
1. Equipotential surfaces of a positive point charge.



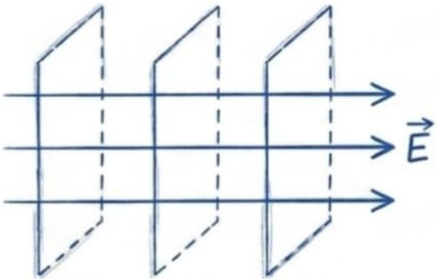
2. Equipotential surface of 2 equal & opposite charges or an electric dipole.



3. Equipotential surfaces of 2 equal & like charges:

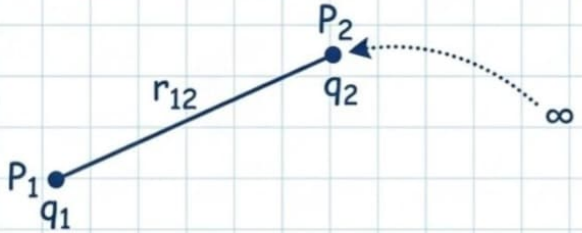


4. Equipotential surface of a uniform electric field:



Electric potential energy of a system of point charges

For 2 point charges



Work done to carry q_1 alone from ∞ at P_1 isolated, $W = 0$

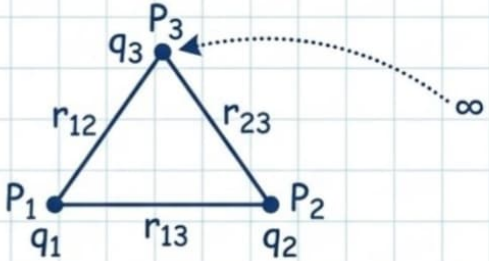
Now, when q_1 is at P_1, for bringing q_2 from ∞ to P_2, the Potential due to q_1 is:

$$V_1 = \frac{k\,q_1}{r_{12}}$$

Now, work done to carry q_2 from ∞ to P_2:

$$W_2 = q_2[V_1 - V_\infty]$$

For 3 point charges



Now in order to bring q_3 from ∞ to P_3, Potential at P_3 due to q_1 & q_2:

$$V_3 = \frac{k\,q_1}{r_{13}} + \frac{k\,q_2}{r_{23}}$$

Now, to bring q_3 from ∞ to P_3:

$$W_3 = q_3[V_3 - V_\infty] \quad \text{or}$$

$$W_3 = kq_3\left[\frac{q_1}{r_{13}} + \frac{q_2}{r_{23}}\right]$$

Total Potential Energy (U)

Now, in order to bring all 3 charges together, total work done is stored in form of Potential energy (U).

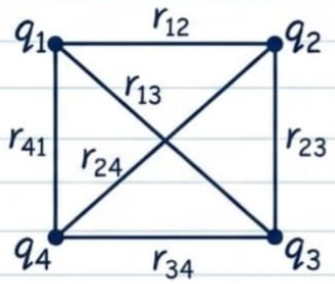
$$\therefore U = W_1 + W_2 + W_3$$

$$= 0 + \frac{k \cdot q_1 q_2}{r_{12}} + \frac{k q_1 q_3}{r_{13}} + \frac{k q_2 q_3}{r_{23}}$$

$$\boxed{\therefore U = \frac{1}{4\pi\varepsilon_0}\left[\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}}\right]}$$

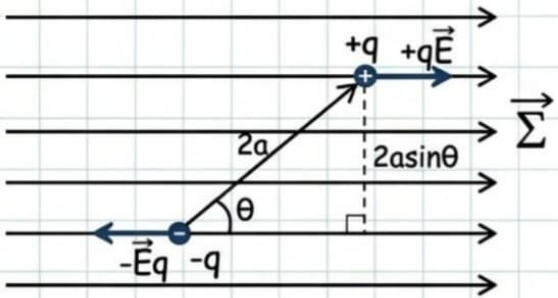
Notice the pattern! The total energy is simply the sum of the potential energy of every unique pair of charges in the system.

Extension to 4 point charges



$$\therefore U = \frac{1}{4\pi\varepsilon_0}\left[\frac{q_1 q_2}{r_{12}} + \frac{q_2 q_3}{r_{23}} + \frac{q_3 q_4}{r_{34}} + \frac{q_4 q_1}{r_{41}} + \frac{q_2 q_4}{r_{24}} + \frac{q_1 q_3}{r_{13}}\right]$$

Potential Energy of a dipole in a uniform electric field



The uniform electric field

Here, in uniform electric field, no translational force act on a dipole but a torque acts on it given by,

$$\vec{\tau} = \vec{P} \times \vec{\Sigma}$$

$$\text{or} \quad \tau = PE\sin\theta$$

In order to rotate dipole in opposite order of its pre-existing torque direction by angle $d\theta$. So, work done for it is :

$$dW = \tau \cdot d\theta \Rightarrow dW = PE\sin\theta \cdot d\theta$$

In order to rotate dipole from θ_1 to θ_2,

$$\int dW = \int_{\theta_1}^{\theta_2} PE\sin\theta \cdot d\theta$$

$$W = PE\int_{\theta_1}^{\theta_2} \sin\theta \cdot d\theta$$

$$W = PE\left[-\cos\theta\right]_{\theta_1}^{\theta_2}$$

$$W = -PE\left[\cos\theta_2 - \cos\theta_1\right]$$

Dipole Equilibrium States

If $\theta_1 = 90°$, $\theta_2 = \theta$

$\therefore W = PE\,[\cos 90° - \cos\theta]$

or $W = PE\,[0 - \cos\theta]$

This work stores in the form of [U],

$$\boxed{U = -PE\cos\theta}$$

A dipole acts like a compass needle. It wants to align with the field (θ=0°, lowest energy). Facing perfectly backwards (θ=180°) takes maximum energy and is highly unstable.

Unstable

* $\theta = 180°$

At $\theta = 180°$, $\cos 180° = -1$

$\therefore U = -PE \cos 180° = -PE(-1) = +PE$

$\therefore U_{max} = +PE$

Unstable position

Stable

* $\theta = 0°$

At $\theta = 0°$, $\cos 0° = 1$

$\therefore U = -PE \cos 0° = -PE(+1) = -PE$

Stable position

Capacitance

For an insulated conductor,

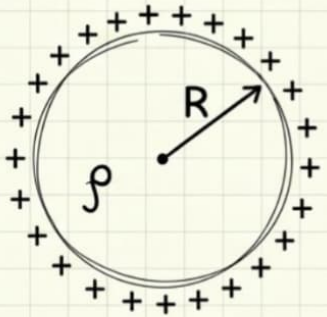
$$Q \propto V$$

or $$Q = C \cdot V$$

Where C is the constant of proportionality and called as capacitance.

$\therefore C = \frac{Q}{V}$ --- Its SI is $\frac{A}{V}$ or farad (F).

✳ Capacitance of Spherical conductor



In this sphere,

Potential on surface of sphere: $V = \frac{kq}{R}$

Let, $Q = q$ i.e. charge on sphere

$$\therefore C = \frac{Q}{V} = \frac{q \cdot R}{kq} = \frac{R}{k} = 4\pi\varepsilon_0 R$$

$$\boxed{C = 4\pi\varepsilon_0 R \quad \text{or} \quad C \propto R}$$

Capacitance

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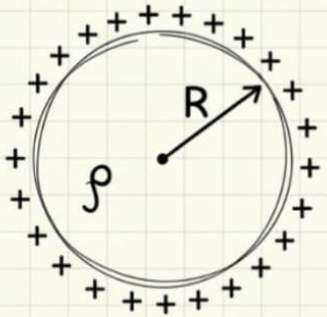

In this sphere,

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Let, $Q = q$ i.e. charge on sphere

$$\therefore C = \frac{Q}{V} = \frac{q \cdot R}{kq} = \frac{R}{k} = 4\pi\varepsilon_0 R$$

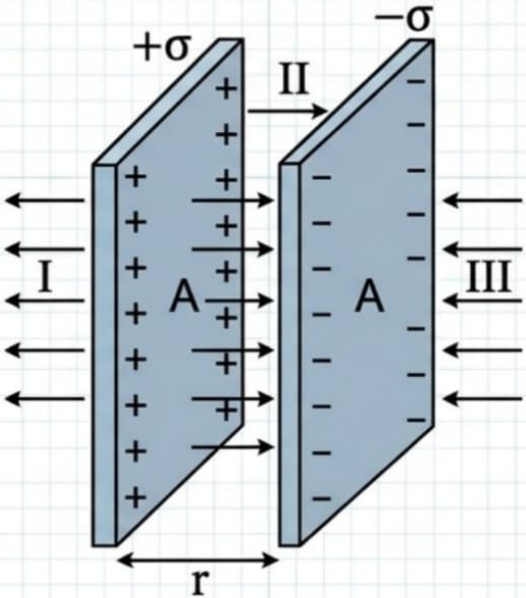
$$\boxed{C = 4\pi\varepsilon_0 R \quad \text{or} \quad C \propto R}$$

Parallel Plate Capacitors

The simplest & most widely used **capacitor** is **parallel plate capacitor**. It consists of 2 large plane parallel conducting plates, separated by a small distance.

- Let A = area of each plate.
- r = distance b/w 2 plates.
- $\pm\sigma$ = uniform surface charge densities on 2 plates.

$\therefore$ $\pm Q = \pm\sigma \cdot A$ = total charge on each plate.



Now, As EF due to ∞ plane sheet = $\frac{\sigma}{2\epsilon_0}$

Now, In regions I & III the net field will be 0, since, equal & opposite fields.

But, between || plates,

$$E = 2 \cdot \frac{\sigma}{2\epsilon_0} = \frac{\sigma}{\epsilon_0}$$

$$\boxed{E = \frac{\sigma}{\epsilon_0}}$$

Parallel plate capacitance

Now, potential difference between plates,

$$V = E \cdot r \quad \text{or} \quad V = \frac{\sigma}{\varepsilon_0} \cdot r \quad \text{--- (I)}$$

Now, $Q = \sigma \cdot A$ --- (II)

$$C = \frac{Q}{V} \quad \text{From (I) \& (II),} \quad C = \frac{\sigma A}{\frac{\sigma r}{\varepsilon_0}} = \frac{A\varepsilon_0}{r}$$

$$\boxed{\Rightarrow C \propto A, \quad C \propto \frac{1}{r}}$$

$$\frac{C_1}{C_2} = \frac{A_1}{A_2} \quad \& \quad \frac{C_1}{C_2} = \frac{r_2}{r_1}$$

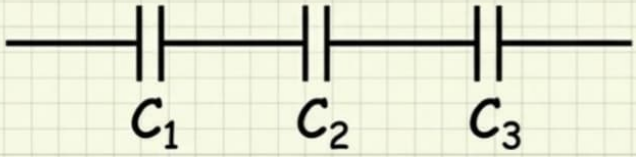
If medium between plates is other than vacuum, then dielectric constant (K) of new medium,

$$C' = \frac{K \cdot \varepsilon_0 \cdot A}{r} \quad \text{or} \quad C' = K \cdot C$$

Notation quirk!
The student uses 'r' for the distance between the plates here, whereas standard textbooks usually use 'd'. The logic remains perfectly identical.

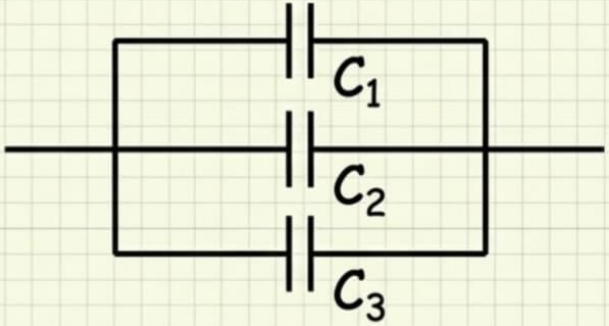
Capacitor Combinations

Series combination



$C_{net} = \frac{C}{n}$ In series,

Parallel combination

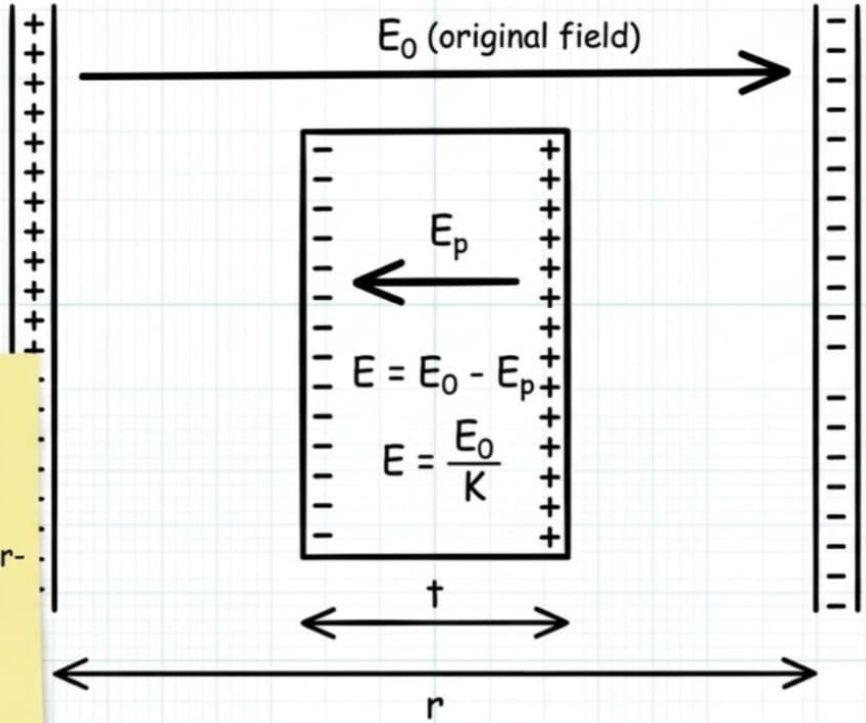


$C_{net} = C_1 + C_2 + C_3$

If all the capacitors are identical each of value [C] & (n) in number.

$C_{net} = C \cdot n$ (For parallel)

Parallel plate capacitor with a dielectric slab inside it



This diagram is crucial. The dielectric doesn't block the field entirely, but its internal induced charges create a counter-field (E_p) that weakens the net field inside the slab to E_0/K.

Dielectric Slab Derivation

Now, in between plallel plates, $E_0 = \frac{2\cdot\sigma}{2\varepsilon_0} \Rightarrow E_0 = \frac{\sigma}{\varepsilon_0}$

Also, in slab, $E = \frac{E_0}{K}$

Now, potential difference b/w plates,

$$V = E_0(r-t) + E(t)$$

$$\text{or } V = E_0(r-t) + \frac{E_0(t)}{K}$$

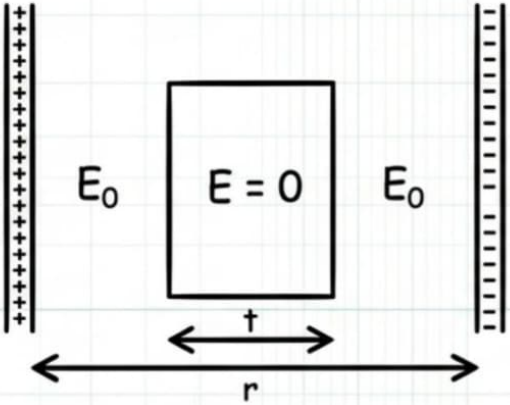
$$\therefore V = \frac{\sigma}{\varepsilon_0}\left[r - t + \frac{t}{K}\right]$$

As, Q on plate is $Q = \sigma\cdot A$

$$C = \frac{Q}{V} = \frac{\sigma\cdot A}{\frac{\sigma}{\varepsilon_0}\left[r - t + \frac{t}{K}\right]}$$

$$\boxed{C = \frac{A\varepsilon_0}{r - t + \frac{t}{K}}}$$

Parallel plate capacitor with a conducting slab inside it



Dielectric constant of metal conductor is ∞

$$\therefore C = \frac{\varepsilon_0 A}{r - t + \frac{t}{\infty}}$$

$$\boxed{\therefore C_{\text{conducting slab}} = \frac{\varepsilon_0 A}{r - t}}$$

If there is no slab inside plates,

$$t = 0$$

$$\therefore C = \frac{\varepsilon_0 A}{r - 0 + \frac{0}{K}} = \frac{\varepsilon_0 A}{r}$$

Sanity Check:
Setting t=0 collapses our complex formula back down to the standard parallel plate equation ($C=\varepsilon_0 A/r$). Physics works perfectly!